Iowa State University

Electrical And Computer Engineering

EE 230

LAB 5 REPORT

October 8, 2025

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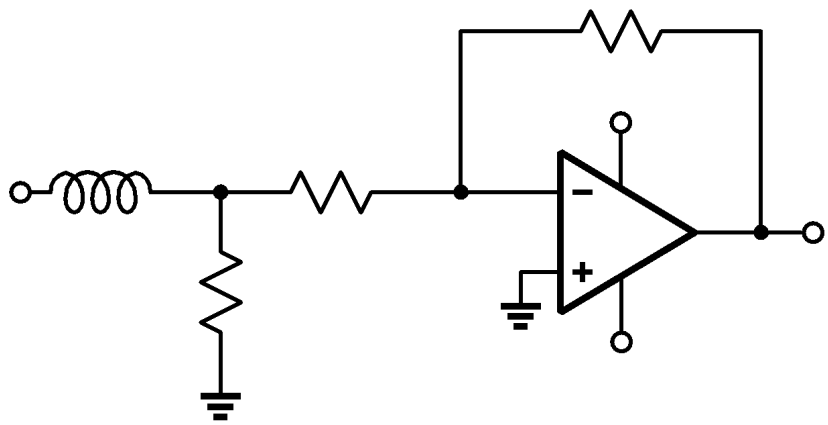
Introduction

In this lab we analyzed a circuit that was given to us along with the respective values. It was also given what output the circuit was SUPPOSED to have. We measured things like magnitude and phase to figure out what the output truly was. It was wrong. We adjusted values of the resistor R2 and R3 to make the circuit output correctly.

A Low-Pass Filter with Gain – Measurement & Analysis



The table below lists the nominal and measured component values, with units:

Component	Nominal Value	Measured Value
L	47mH	46.98072 mH
R_S	0 Ω	134.4110 Ω
R_1	330 Ω	328.3626 Ω
R_2	1 kΩ	1.010962 kΩ
R_3	2.2 kΩ	2.166356 kΩ

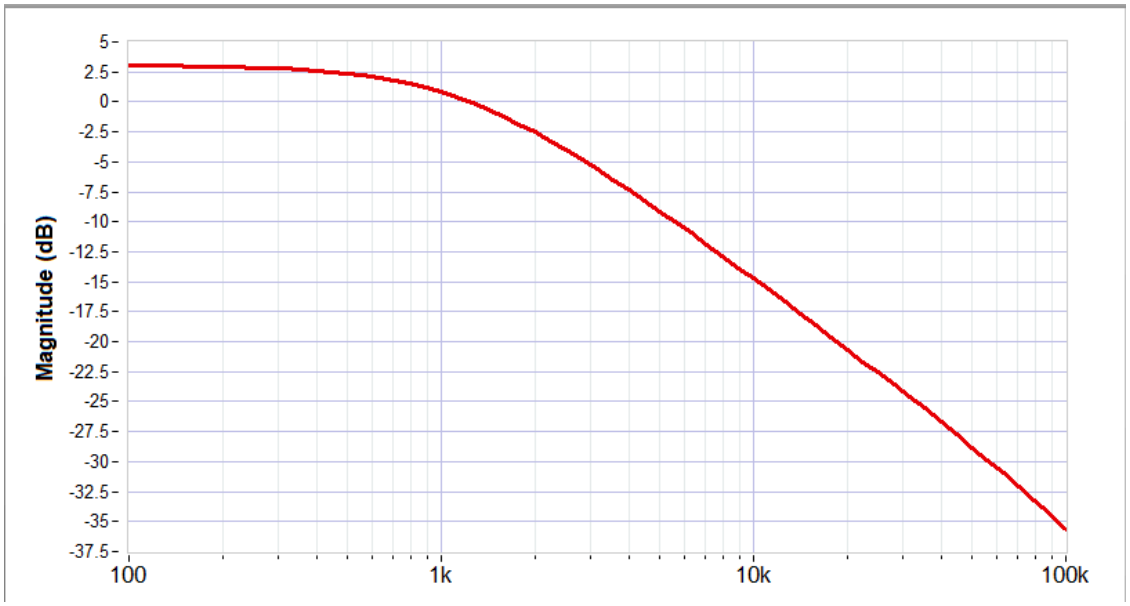


***Figure 1**: Measured magnitude response obtained from the Loopinator*

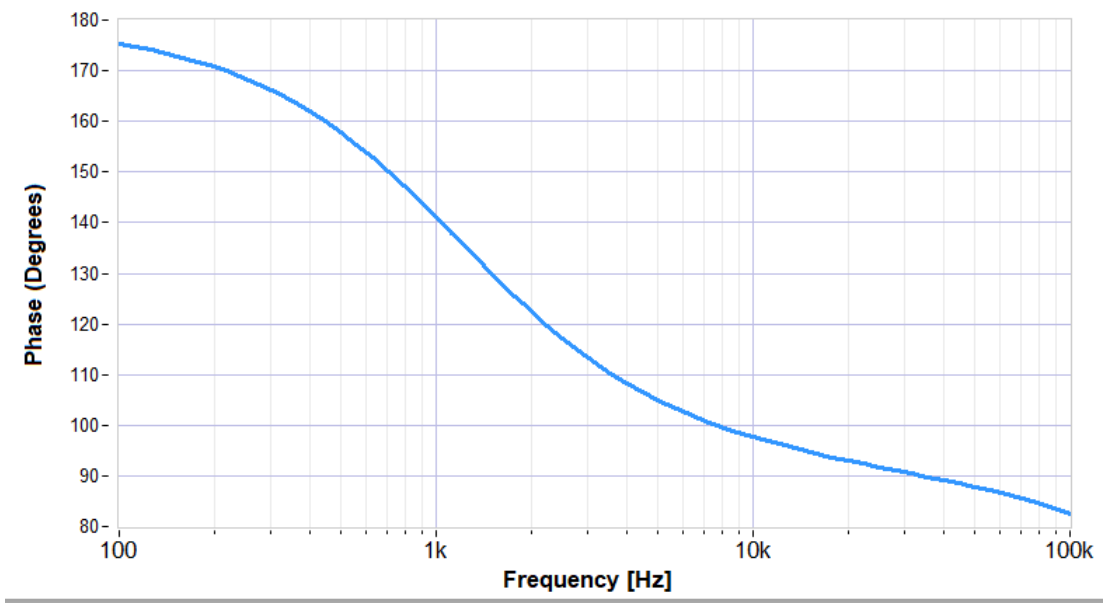
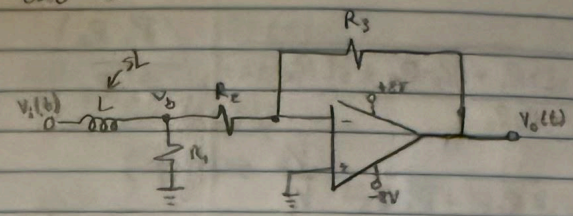


***Figure 2**: Measured phase response obtained from the Loopinator.*

The table below lists the measured cutoff frequency in Hz and the measured DC gain in V/V:

Quantity	Measured Value
f_c	1259 Hz
DC Gain	1.41

Lab 5



$L = 47\,mH$
$R_1 = 330\,\Omega$
$R_2 = 1\,k\Omega$
$R_3 = 2.2\,k\Omega$

Transfer function

$$\frac{0-V_b}{R_2} + \frac{0-V_o}{R_3} = 0 \Rightarrow \frac{-V_b}{R_2} - \frac{V_o}{R_3} = 0 \Rightarrow V_o = \frac{-R_3}{R_2} V_b$$

KCL at V_b

$$\frac{V_b - V_i}{sL} + \frac{V_b}{R_1} + \frac{V_b}{R_2} = 0 \Rightarrow \frac{V_i - V_b}{sL} = V_b\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \Rightarrow V_i = V_b\left[1 + sL\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\right]$$

$$\frac{V_b}{V_i} = \frac{1}{1 + sL\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} = \frac{1}{1 + s\frac{L}{R_1 \| R_2}}$$

$$H(s) = \frac{V_o}{V_i} = \frac{V_o}{V_b} \cdot \frac{V_b}{V_i} = -\frac{R_3}{R_2} \frac{R_1 \| R_2}{sL + R_1 \| R_2} = \frac{-R_3 R_1}{R_1 + R_2} \frac{1}{sL + \frac{R_1 R_2}{R_1 + R_2}}$$

$$|H(j\omega)| = \frac{|R_3/R_2|}{\sqrt{1 + \left(\frac{\omega}{\omega_p}\right)^2}} \qquad -\frac{R_3}{R_2} \frac{1}{1 + \frac{s}{\omega_p}}$$

With parasitic

at V_b: $$\frac{V_b - V_i}{sL + R_s} + \frac{V_b}{R_1} + \frac{V_b}{R_2} = 0 \rightarrow (sL + R_s) V_b \left(\frac{1}{sL + R_s} + \frac{1}{R_1} + \frac{1}{R_2}\right) = V_i$$

$$V_b\left(1 + \frac{sL + R_s}{R_1} + \frac{sL + R_s}{R_2}\right) = V_i \quad \Big| \quad \frac{V_b}{V_i} = \frac{1}{\left(1 + \frac{sL + R_s}{R_1} + \frac{sL + R_s}{R_2}\right)}$$

• in denominator:

$$\left(1 + \left(\frac{sL + R_s}{R_1}\right)\left(\frac{R_2}{R_2}\right) + \left(\frac{sL + R_s}{R_2}\right)\left(\frac{R_1}{R_1}\right)\right) = 1 + \frac{sL \cdot R_2 + R_s \cdot R_2 + sL \cdot R_1 + R_s \cdot R_1}{R_1 R_2}$$

Continue on next page

$$\left(\frac{1}{1+\left(\frac{R_s\cdot R_1 + sL\cdot R_1 + R_s\cdot R_2 + sLR_2}{R_1R_2}\right)}\right)\left(\frac{R_1\cdot R_2}{R_1\cdot R_2}\right)$$

$$= \frac{R_1\cdot R_2}{R_1R_2 + sLR_1 + sLR_2 + R_sR_1 + R_sR_2} = \frac{V_b}{V_i} = H_1(s)$$

$$H(s) = H_1(s)\cdot H_2(s), \quad H_2(s) = \frac{V_o}{V_b} = \frac{-R_3}{R_2}$$

$$\left(\frac{R_1\cdot \cancel{R_2}}{R_1R_2 + sLR_1 + sLR_2 + R_sR_1 + R_sR_2}\right)\left(\frac{-R_3}{\cancel{R_2}}\right)$$

$$H(s) = \frac{-R_1\cdot R_3}{R_1\cdot R_2 + sLR_1 + sLR_2 + R_sR_1 + R_sR_2}$$ Expanded version

$$H(s) = \frac{-R_1\cdot R_3}{R_1\cdot R_2 + (R_1+R_2)\cdot(sL+R_s)}$$ *compacted denominator

$$\frac{-R_1\cdot R_3}{R_1\cdot R_2 + (R_1+R_2)\cdot(sL+R_s)}$$

Standard form $= G_0\frac{\omega_c}{s+\omega_c}$

$$R_1R_2 + (R_1+R_2)(sL+R_s) \Rightarrow R_1R_2 + R_1sL + R_1R_s + R_2sL + R_2R_s$$

$$\Rightarrow R_1R_2 + R_1R_s + R_2R_s + s(LR_1 + LR_2)$$

$$\Rightarrow \frac{R_1R_2 + R_1R_s + R_2R_s}{LR_1 + LR_2} + s$$

$$\frac{-R_1R_3}{\frac{R_1R_2 + R_1R_s + R_2R_s}{LR_1 + LR_2} + s} \Rightarrow$$

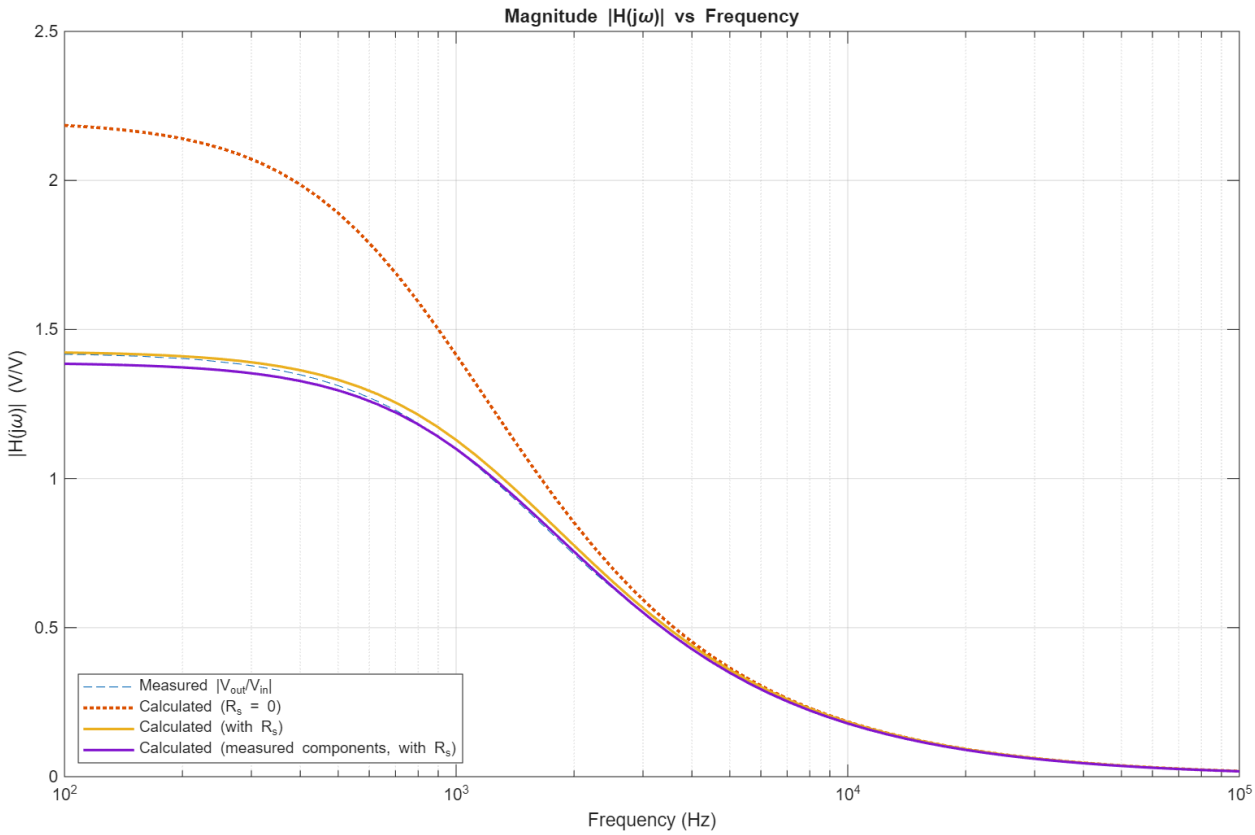


Figure 3*: Plot of measured and calculated magnitude response. The calculated response includes one trace using nominal component values (with $R_S = 0\ \Omega$) and another trace using measured component values (with $R_S \neq 0\ \Omega$).*

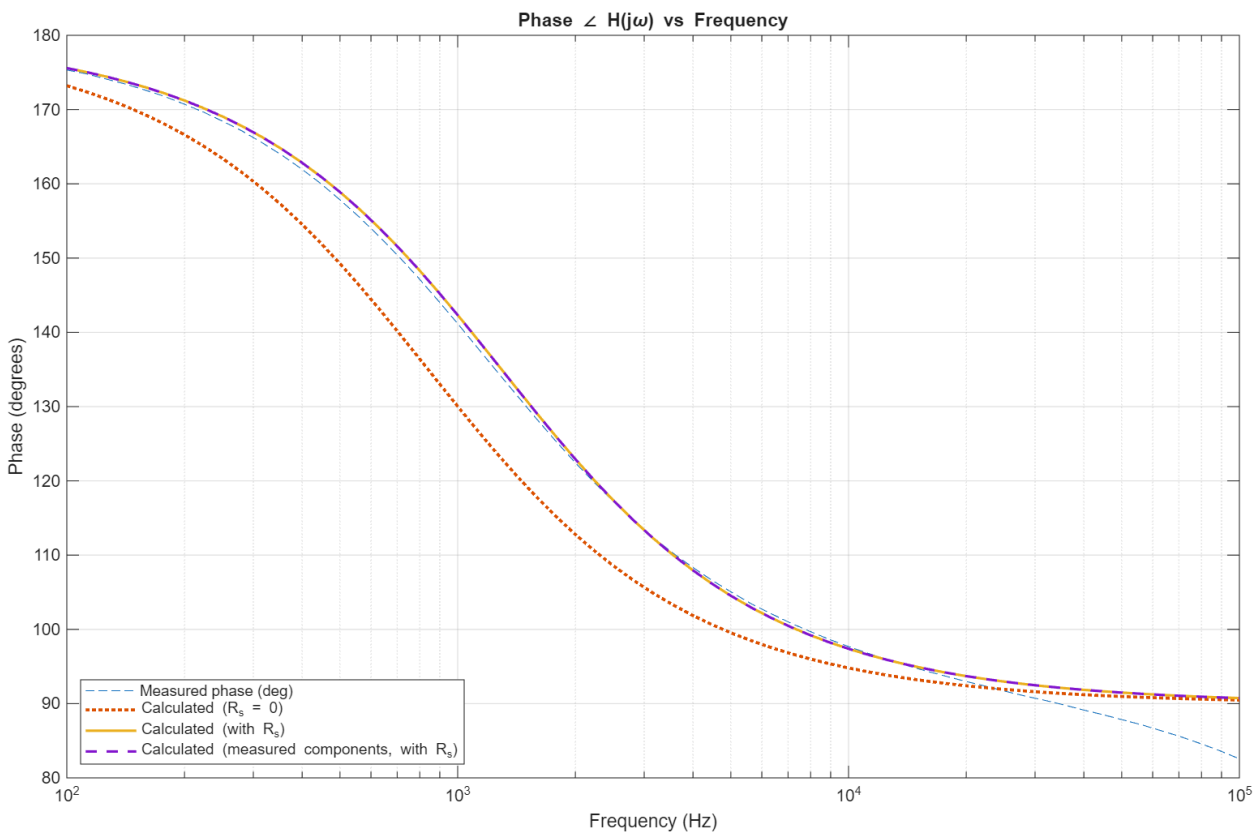


Figure 4*: Plot of measured and calculated phase response. The calculated response includes one trace using nominal component values (with $R_S = 0\ \Omega$) and another trace using measured component values (with $R_S \neq 0\ \Omega$).*

The measured response differs from the ideal response mainly because real components are not ideal. The inductor used in the circuit has a parasitic resistance which reduces the effective feedback in the amplifier network. This lowers the DC gain from the expected −2.2 V/V and shifts the cutoff frequency from 1.1 kHz.

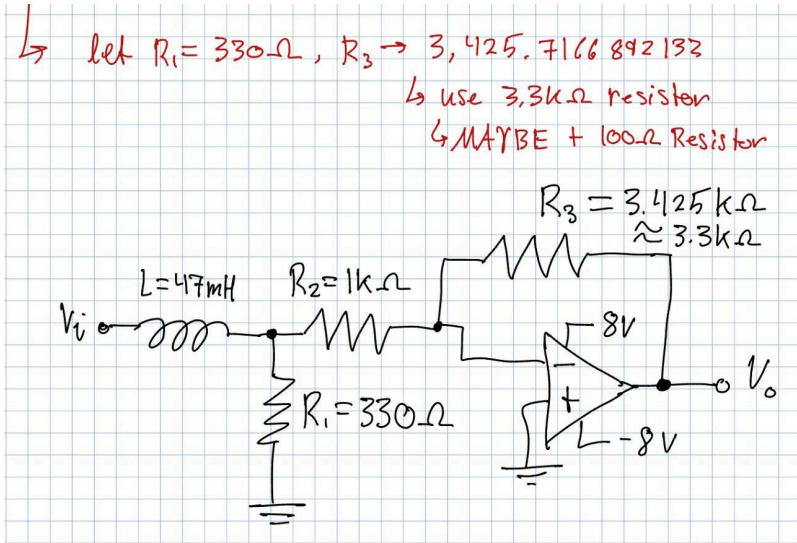


***Figure 5**: Schematic of newly designed low-pass filter with gain.*

The table below lists the nominal and measured component values, with units, for your newly design circuit.

Component	Nominal Value	Measured Value
L	47 mH	46.98072 mH
R_S	0 Ω	134.4110 Ω
R1	330 Ω	325.69
R2	1 kΩ	1.010962 kΩ
R3	3.3 kΩ	3.2547k

(if R1 = 330 Ω, R3 = 3.425 kΩ)

We wanted to keep one of the resistor values the same because a TA told us we would have to do our calculations that way. It would be way too complicated if we didn't. So we kept R_2 at 1kΩ and solved for R1 and R3. We kept everything else the same because we realistically cannot change the ground, the wires*, the inductor, breadboard, and op amp with enough control to get the output that we want. So we change the resistor(s) values. You can see the algebra and Symbolic Transfer Function at the end of this Lab Report within the calculations.

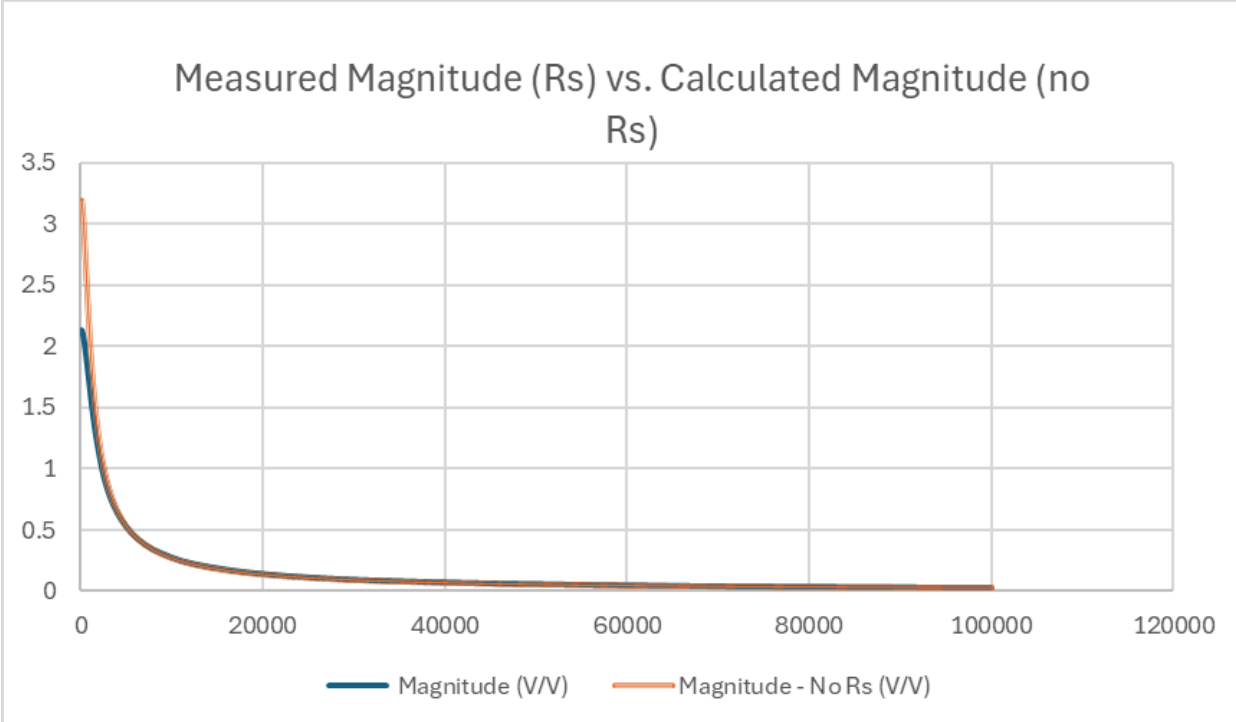


***Figure 6**: Plot of measured and calculated magnitude response for the re-designed filter. The calculated response includes one trace using nominal component values (with $R_S = 0\ \Omega$) and another trace using measured component values (with $R_S \neq 0\ \Omega$).*

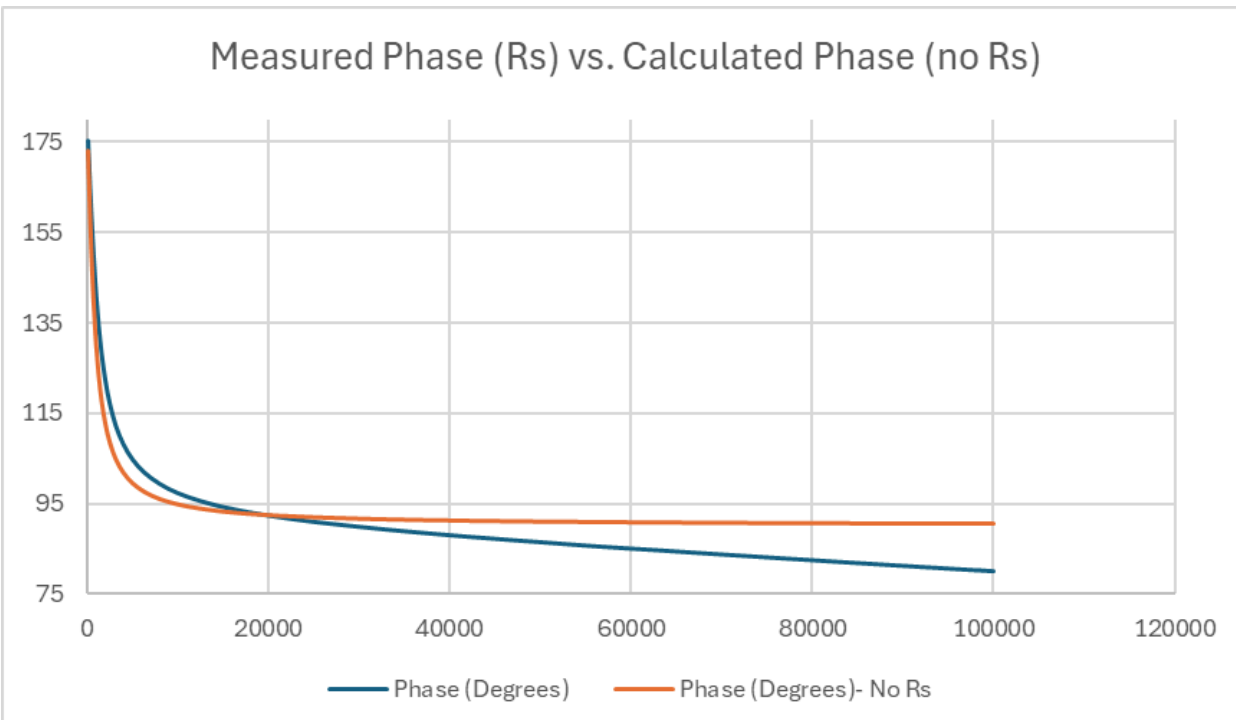


***Figure 7**: Plot of measured and calculated phase response for the re-designed filter. The calculated response includes one trace using nominal component values (with $R_S = 0\ \Omega$) and another trace using measured component values (with $R_S \neq 0\ \Omega$).*

The table below lists the measured cutoff frequency in Hz and the measured DC gain in V/V for the re-designed filter:

Quantity	Measured Value	Error (%)
f_c	1129	
DC Gain	2.137	

Conclusion

In this lab we used the oscilloscope for the loopinator. We used it to verify the output of the given circuit with its given values. It turned out to be way wrong.

We calculated the transfer function of the ideal and non ideal circuit, with the non ideal circuit accounting for the parasitic resistance of the inductor while the ideal transfer function did not.

The non-ideal transfer function was used. We got it into standard form, allowing us to point out ω_o which then allowed us to derive G_o both symbolically. Using those equations we were able to set 1 of the resistor values to a somewhat arbitrary constant, set the equation equal to our desired G_o , and derive an equation with two unknown resistor values. We then tried numerous resistor values for one to get a resistor value for the other in which both resistor values were close to that of resistors in our lab kits.

We then measured the resistors and swapped out the resistors in the original circuit for our calculated correct resistor values. Doing so allowed us to verify our new/fixed circuit using the loopinator where we were able to identify near 2.2 V/V gain and a near 180 degree phase when the frequency was 100 Hz. We were able to mentally extrapolate the data to ~0 Hz which got our values even closer to 2.2 V/V and 180 degrees. So the resistor values we chose were correct.

Calculations

$$H(s) = \frac{-R_1 R_3}{R_1 R_2 + R_3 R_1 + R_3 R_2 + sLR_1 + sLR_2} = -2.2\ V/V$$

$j\omega \rightarrow \omega$ / 1.1 kHz $\rightarrow$ substitute in $j(2\pi f_c)$ for s

↳ let $R_1 = 330\Omega$, $R_3 \rightarrow 3,425.7166892133$

↳ use $3.3k\Omega$ resistor

↳ MAYBE + 100Ω Resistor